



Synthesis and evaluation of acrylate resins suspending indole derivative structure in the side chain for marine antifouling

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ABSTRACT

A novel indole derivative (N-(1H-2-phenyl-indole-3-ylmethyl) acrylamide, NPI) synthesized by a Friedel-Crafts alkylation reaction was identified using IR spectroscopy, ¹H NMR, ¹³C NMR and elemental analysis. The inhibitory effect of this novel indole derivative on bacteria and marine algae was studied. The results showed that the inhibition ratios of the indole derivative against *Escherichia coli* and *Staphylococcus aureus* were 95.93% and 94.91%, respectively, and the indole derivative possessed prominent inhibitory activity against *Phaeodactylum tricornutum*, *Nitzschia closterium* and *Skeletonema costatum*. These findings indicate that the indole derivative has high biological activity. Subsequently, the indole derivative was introduced to acrylate resins by free-radical polymerization. The resulting acrylate resins were subjected to self-polishing, anti-algal and antifouling test, the results of which indicated that acrylate resins containing the synthesized indole derivative could exhibit significant antifouling properties because of the combination of the biofouling resistance of the indole derivative and the self-polishing properties of acrylate. This work provides an academic foundation for studying environmentally friendly and highly efficient antifouling coatings.

1. Introduction

Since humans first participated in marine development, marine biofouling has resulted in numerous problems for marine exploitation activities, such as reduced ship speed, increased fuel consumption [1,2], blocked pipes, accelerated corrosion and increased greenhouse gas emissions [3]. Thus, marine biofouling causes considerable economic losses [4,5]. With the continuous exploitation of marine environments, the marine antifouling requirements for marine instruments as well as civil and military ships are increasing. In particular, large warships require a longer antifouling period than that of other types of ships; the Naval Sea Systems Command set up a program in 2002 that demanded an antifouling period of up to 12 years [6]. The most effective method of addressing biofouling issues is antifouling coatings that prevent the surface attachment of fouling organisms. Due to the effective antifouling property of organotin-based coatings [7], they have been applied extensively since the early 1970s [8]. However, the International Maritime Organization has banned organotin antifouling paints globally due to the serious damage to the marine ecological environment [9–11]. Therefore, research and development of environment-friendly antifouling coatings is imperative [12,13]. To create more environmentally friendly and effective marine antifouling

coatings, natural antifouling agents have become a popular research topic, mainly including terpenes, nitrogen-containing compounds, phenols, steroids, etc. For example, the indole derivatives 2,5,6-tri-bromo-1-methylgramine derived from the bryozoan *Zoobotryon pellucidum* [14] and indole-3-carboxaldehyde derived from ascidians [15] have been demonstrated to act as antifouling agents, exhibiting significant inhibition of surface attachment by marine organisms [16]. Indole derivatives have proved to be promising natural antifouling agents because of their easy synthesis, simple structure and high antifouling activity.

To find new compounds with enhanced antifouling activity, activity studies on analogues of natural indole-containing antifouling compounds have been reported in recent years. Furthermore, the environmental safety of these compounds has been extensively studied. Wang et al. studied the inhibitory effects of diindol-3-ylmethanes against larval settlement of *B. amphitrite*. The results showed diindol-3-ylmethanes could be considered as low-toxicity antifoulants since their LC₅₀/EC₅₀ ratios were larger than 15 [17]. Pérez et al. reported that bufotenine showed significant antifouling activity against three types of marine biofilm bacteria in the Microtox assay despite being nontoxic [16]. In addition, many studies have indicated that indole derivatives could be degraded [18–20]. For example, Yin et al. revealed that 3-

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methylindole could be rapidly degraded by indigenous microorganisms found in mangrove sediment. The above studies indicate that indole derivatives not only possess high activity but also are relatively environment-friendly.

Early in our study, a series of novel antifoulant analogues of natural products containing acylamino groups at the 3-position of the indole ring were synthesized via a Friedel-Crafts alkylation reaction. The inhibitory effect of the novel indole derivatives on bacteria and marine algae was studied, and the results showed that indole derivative exhibited excellent inhibitory activity. However, the effect of antifouling coatings with added indole derivatives used as antifoulants is poor. This inadequacy is because the release rates of the indole derivatives from the antifouling coatings were uncontrollable. A feasible solution is to introduce indole derivatives into macromolecules to ensure long-term antifouling effects. Considering the film formers being extensively used in antifouling coatings at present, acrylate resins that have self-polishing properties were selected. Moreover, given a previous study of the structure-activity relationship of indole derivatives [21], the indole derivative containing a carbon-carbon double bond and another phenyl was purposefully synthesized. Acrylate resins suspending the indole derivative structure in their side chain were prepared by the free-radical polymerization, and dynamic simulation measurements, static immersion measurements, algal inhibition testing and practical panel testing in a marine environment were performed to study their self-polishing and antifouling properties.

2. Material and methods

2.1. Instruments and reagents

This study used the following for the investigation: MLS-3750 autoclave sterilizer (Sanyo, Japan), HZQ-F160 incubator shaker (Harbin Donglian company), U-2800 ultraviolet spectrophotometer (Hitachi, Japan), MLR-350H Versatile Environmental Test Chamber (Sanyo, Japan), LEXT OLS4000 laser scanning confocal microscope (Olympus, Japan). IR spectra were recorded for KBr disc on a Tensor 27 Fourier Transform Infrared Spectrometer (Bruker, Germany). ^1H NMR and ^{13}C NMR spectra were recorded with AVANCE III (600 MHz) instrument (Bruker, Switzerland) in $\text{DMSO}-d_6$ as the solvent, using TMS as an internal standard, and chemical shifts were indicated in parts per million (δ). 2-phenyl-indole, N-methylol acrylamide, chlorothalonil and acrylic monomers were obtained from Shanghai Aladdin Bio-Chem Technology Co., LTD, absolute ethanol, sulfuric acid, acetone and azodiisobutyronitrile were purchased from Sinopharm Chemical Reagent Co., Ltd. All reagents were used as received.

2.2. Synthesis of indole derivative

To a stirred solution of 2-phenyl-indole (19.32 g, 0.1 mol), N-methylol acrylamide (12.13 g, 0.12 mol) in dichloromethane (50 mL) was added dropwise a suspension of anhydrous aluminum chloride (1.34 g, 10 mmol) in dichloromethane (20 mL) in an ice bath. The mixture was gradually warmed to room temperature after 2 h of stirring in an ice bath. After another 48 h of stirring at room temperature, the reaction mixture was added to the ice diluted hydrochloric acid solution (1.1 g, 30 mmol). The organic layer was separated, washed with deionized water, dried (MgSO_4), and concentrated in vacuo to obtain the crude residue. The crude residue was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 1 : 1) to afford the indole derivative. The indole derivative was analyzed by IR, ^1H NMR, ^{13}C NMR and elemental analysis. The structure of indole derivative is shown in Fig. 1.

N-(1H-2-phenyl-indole-3-methyl) acrylamide: IR (KBr) ν : 1412, 1452, 1534, 1616, 1655, 2926, 3288, 3382 cm^{-1} ; ^1H NMR ($\text{DMSO}-d_6$, 600 MHz) δ : 4.61 (t, 2H, CH_2), 5.61 (m, 1H, $\text{CH}_2=\text{C}$), 6.20 (m, 1H, $\text{CH}_2=\text{C}$), 6.31 (m, 1H, $\text{CH}=\text{C}$), 7.05 (m, 1H, PhH), 7.15 (m, 1H, PhH),

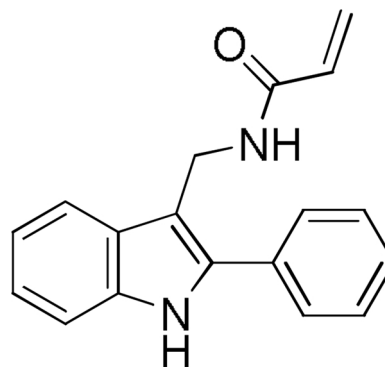


Fig. 1. Structure of indole derivative (N-(1H-2-phenyl-indole-3-methyl) acrylamide, NPI).

7.43 (t, 2H, PhH), 7.54 (t, 2H, PhH), 7.62 (d, 1H, PhH), 7.71 (d, 1H, PhH), 7.72 (d, 1H, PhH), 8.43 (t, 1H, CH_2NH), 11.3 (s, 1H, NH); ^{13}C NMR ($\text{DMSO}-d_6$, 150 MHz) δ : 34.17, 108.77, 111.71, 119.42, 119.60, 122.32, 125.60, 128.22, 128.69, 128.84, 129.25, 132.23, 132.67, 136.40, 136.43, 164.87; Anal. Calcd (%) for $\text{C}_{18}\text{H}_{16}\text{N}_2\text{O}$: C 78.24, H 5.84, N 10.14; found: C 78.91, H 6.62, N 9.87. The IR, ^1H NMR and ^{13}C NMR spectra of indole derivative are shown in the supplementary material (Fig. A.1).

2.3. Antibacterial activity test of indole derivative

E. coli and *S. aureus* inoculum were provided by the microbiology lab of the College of Marine Life Science at Ocean University of China. Sterilization treatment is necessary before the antibacterial activity test. Slant preservation bacteria of *S. aureus* and *E. coli* were cultivated for 20 h at $37 \pm 1^\circ\text{C}$, and the strains were scraped by an inoculation loop from the slant culture medium to sterile liquid culture medium (nutrient broth). Then, the bacterial suspension was cultured for 3 h. The biomass concentration of the bacterial suspension was 10^8 CFU per mL. The indole derivative and chlorothalonil (0.5 mg/mL) were dissolved in acetone, and acetone served as a blank. The sterile liquid culture medium (15 mL), test samples (1 mL) and bacterial suspension (200 μL) were placed in a test tube, and the sample solution for the test was cultivated for 18 h in an incubator shaker. Normal saline was added to the sample solution at a certain factor for a 10-fold serial dilution, and then the sample solution (200 μL) was uniformly coated on sterile solid medium (nutrient agar) using a triangular coater. The active bacteria were counted after cultivation for 18 h, and antibacterial activity was calculated by the following formula:

$$T = (T_0 - T_i) / T_0 \times 100\% \quad (1)$$

where T is the antibacterial rate, T_0 is the number of bacteria on the control samples, and T_i is the number of bacteria on the experimental samples [22,23]. The above experiment was repeated three times, and the average value was determined. The minimal inhibitory concentration (MIC) test of this derivative for bacteria is provided in the supplementary material.

2.4. Algal inhibition test of indole derivative

Phaeodactylum tricornutum, *Nitzschia closterium*, *Skeletonema costatum* were provided by the algae species lab of the Fisheries College at Ocean University of China. The three kinds of algae were selected because they belong to the diatoms, which are the most common marine plankton type. The maximum absorption wavelengths of the algae were obtained by wavelength scanning from 1000 nm to 500 nm. The algae solution was diluted to various levels during the exponential phase of growth. The absorbance values of various levels at the maximum absorption wavelength were determined. In addition, the concentrations

of the corresponding algal solutions were calculated with a blood cell counting chamber. Standard work curves were obtained with absorbance as the abscissa and concentration as the ordinate.

An algal solution during the exponential phase of growth was diluted to approximately 1×10^5 algal cells per mL using f/2 medium, indole derivative (3 mg/mL) and chlorothalonil (3 mg/mL) dissolved in acetone, and acetone served as a blank. Sterile conical flasks containing diluted algae solution (100 mL) and test samples (0.5 mL) were transferred to a versatile environmental test chamber at a light intensity of 4000 lx and 20 °C with a 12: 12 h light: dark cycle. The absorbance of the algae solution was measured every 24 h to obtain the growth curves with time as the abscissa and concentration as the ordinate [24]. The above experiment was repeated three times, and the average value was determined. The median lethal concentration (LC₅₀) assay of this derivative for algae is provided in the supplementary material.

2.5. Synthesis of acrylate resins suspending the indole derivative structure

A three-necked flask equipped with an agitator and a condenser was put in an oil bath. Stoichiometric xylene and n-butyl alcohol were added as solvents, and particles of azodiisobutyronitrile (AIBN) were dispersed successively in the solvent by an electric mixer. After stirring for 10 min, the stoichiometric acrylic monomers, indole derivative and AIBN were added at 85 °C every 15 min over 2 h of reflux. After that, the resulting solution was kept at a steady temperature for 2 h to obtain the prepolymer (A-NPI-x, x = 0, 5, 10, 15). The stoichiometric prepolymer, Zn(OH)₂, naphthenic acid and mixed solvent were placed in a three-necked flask equipped with an agitator, a condenser and a dewaterer. The system was dehydrated at 110 °C until it became transparent. The acrylate resins containing the indole derivative structure (A-Zn-NPI-x, x = 0, 5, 10, 15) were obtained by anhydrous fractionation and identified by IR and ¹H NMR. The synthetic routes to produce the acrylate resins containing the indole derivative structure are provided in the supplementary material (Fig. A. 2).

Table 1 shows the compositions of the resins with respect to the relative amounts of butyl acrylate (BA), ethyl acrylate (EA), acrylic acid (AA), methacrylic acid (MAA), indole derivative and Zn(OH)₂. The mole fractions of the indole derivative introduced to the acrylate resins are 0%, 5%, 10% and 15%, respectively. Introducing a higher ratio of the indole derivative to the acrylate resins can bring about explosive polymerization, so the introduced amount was limited to 15%.

2.6. Self-polishing performance analysis

The tested samples were coated on an ABS panel (350 mm × 100 mm × 3 mm), and the thickness of the resins was approximately 300 μm (Fig. A. 3a). Fig. A. 3 shows a schematic diagram of the ABS panels and dynamic simulation experiment device for simulated seawater. The conditions under which the dynamic simulation experiment was carried out were as follows: temperature of 15–18 °C, pH 8.2, salinity of 29‰, speed of 80 rpm and a cycle of 12 days. The ABS panels were washed with deionized water at the end of the periodic test and dried to a constant weight (at 35 °C). A laser scanning confocal microscope was used to image the surface morphology variations of the

coatings. The test conditions of the static immersion measurement are the same as those of the above dynamic simulation experiment, and the ABS panels (100 mm × 80 mm × 2 mm) (Fig. A. 3b) were immersed in static seawater. The above experiment was repeated three times, and the average value was determined. A schematic diagram of the ABS panels and dynamic simulation experiment device for simulated seawater are presented in the supplementary material (Fig. A. 3).

2.7. Algal inhibition test of acrylate resins

Algae solutions during the exponential growth phase were diluted to approximately 1×10^5 algal cells per mL using f/2 medium. The resin-coated utensils containing diluted algae solutions were transferred to the versatile environmental test chamber at a light intensity of 4000 lx and 20 °C with a 12: 12 h light: dark cycle. Unhandled utensils containing diluted algae solutions served as a blank. The algae inhibition rates were calculated using the solutions on the seventh day as samples [25,26] by the following formula:

$$S = (S_0 - S_i) / S_0 \times 100\% \quad (2)$$

where S is inhibition rate of algae, S₀ is the absorbance of the control sample, and S_i is the absorbance of the experimental sample [22,23]. The above experiment was repeated three times, and the average value was determined.

2.8. Antifouling test in marine

Compared with the complicated marine environment, only a few marine species are used to assess the antifouling property under the experimental conditions, so an additional antifouling test in a marine environment is very necessary [27]. For this test, the PVC panels were scraped with sandpaper (120 mesh) to increase the roughness and then wiped with xylene. Moreover, acrylate resins were brushed four times on these PVC panels. The prepared panels were dried naturally in a ventilated environment. Ultimately, the thickness of the dry film reached approximately 600 μm. The panels were immersed 1 m deep in Qingdao Harbor, and the marine coating systems were exposed vertically to seawater on July 15, 2018. The panels were removed from the seawater, carefully washed with seawater, and photographed.

3. Results and discussion

3.1. Antibacterial activity test of indole derivative

The antibacterial activity results are presented in Fig. 2. In the control group, abundant bacterial colonies grew on the sterile solid medium, whereas the solid plates with the indole derivative and chlorothalonil had minor colonies of *E. coli* and *S. aureus*. The bacteriostasis rates of the indole derivative and chlorothalonil against *E. coli* are 95.93% and 91.02%, respectively. The bacteriostasis rates of the indole derivative and chlorothalonil against *S. aureus* are 94.91% and 89.35%, respectively. The results demonstrate that the indole derivative possesses better antibacterial properties than chlorothalonil. The difference in bacteriostasis rate between Gram-negative *E. coli* and Gram-positive *S. aureus* could be due to the diversity of the cell wall [28–30], which impacts the penetration of the indole derivative [31]. The peptidoglycan layer in the cell wall of *E. coli* is far smaller than that of *S. aureus* [32], so the indole derivative can more easily cross the cellular membrane of *E. coli* than *S. aureus*. Thus, these discrepancies cause a slight difference in the bacteriostasis rate. Nevertheless, the indole derivative exhibits prominent antibacterial activity. The antibacterial property of the indole derivative serves as the basis for further synthesis of acrylate resins.

Table 1
Compositions of resins with respect to the relative amounts*.

Sample	Monomer					
	BA	EA	AA	MAA	Indole derivative (NPI)	Zn(OH) ₂
A-Zn	4	4	1	1	0 (0%)	2
A-Zn-NPI-5	4	4	1	1	0.526 (5%)	2
A-Zn-NPI-10	4	4	1	1	1.111 (10%)	2
A-Zn-NPI-15	4	4	1	1	1.765 (15%)	2

*The proportion of monomers is molar ratio.

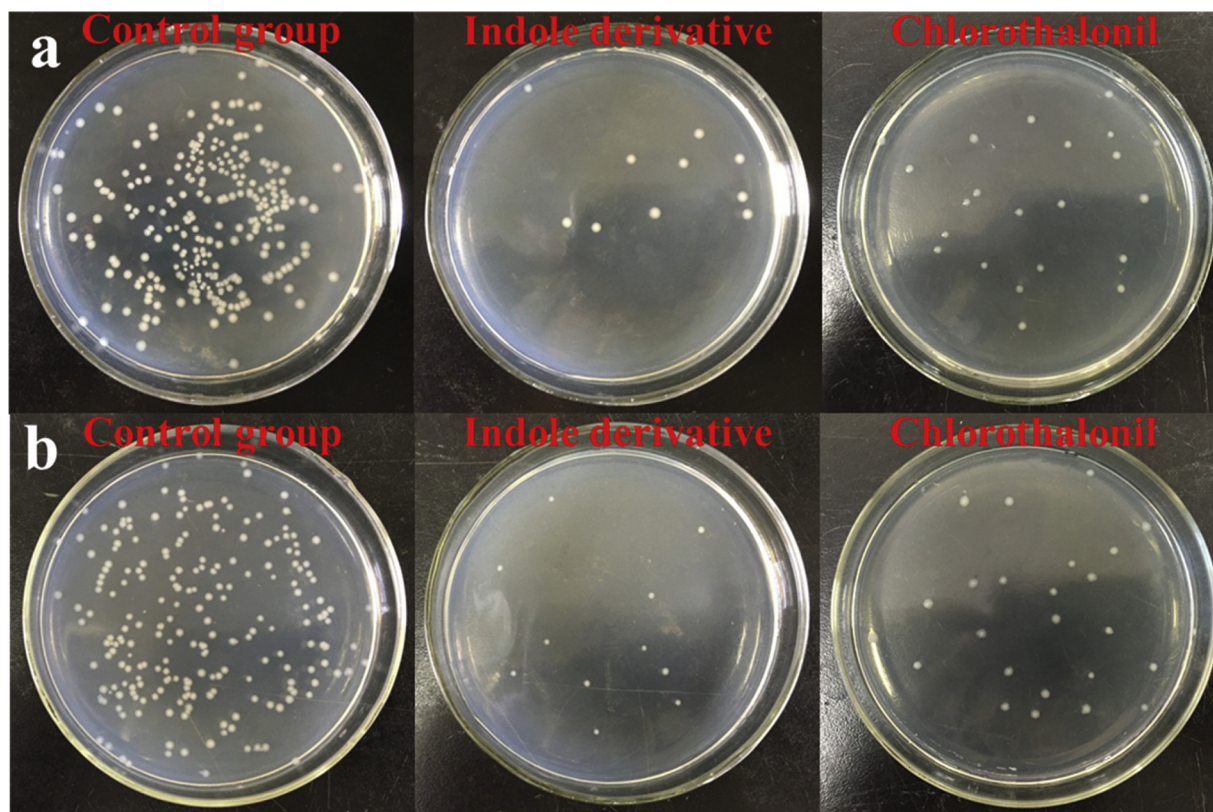


Fig. 2. Antibacterial activity of indole derivative against *E. coli* (a) and *S. aureus* (b).

3.2. Algal inhibition test of indole derivative

Fig. A. 4 shows that the maximum absorption wavelengths of *Phaeodactylum tricornutum*, *Nitzschia Closterium* and *Skeletonema costatum* are 687 nm, 688 nm and 687 nm, respectively. Standard working curves (Fig. A. 5) indicate that the relationship between the absorbance and algae concentration is linear. Chlorothalonil with broad-spectrum bactericidal activity is used as reference [33–35]. The results of the algal inhibition tests of the blank sample, chlorothalonil and the indole derivative are illustrated in Fig. 3. Compared with the blank sample, the other two compounds have obvious inhibitory activity. The indole derivative is superior to chlorothalonil in terms of its inhibitory effects on *Phaeodactylum tricornutum* and *Skeletonema costatum*, while the indole derivative and chlorothalonil have similar inhibitory activity against *Nitzschia closterium*. In brief, the indole derivative could distinctly inhibit the growth of the tested algae. The inhibition mechanism of the indole derivative is speculated to involve the compound interfering with the equilibrium of calcium ions (Ca^{2+}) in algal cells and decreasing the abundance of cellular Ca^{2+} to inhibit the growth of marine algae [36,37]. This mechanism is effective because calcium ions are associated with the growth and development of algae [38,39] and participate in nearly all aspects of algae physiological activities [40,41]. Wavelength scanning and Standard working curves of *Phaeodactylum tricornutum*, *Nitzschia closterium*, *Skeletonema costatum* are provided in the supplementary material (Figs. A. 4 and A. 5).

3.3. Characterizations of acrylate resins suspending indole derivative structure

IR spectroscopy was used to identify various functional groups present in the acrylate resins. The IR spectrum of A-NPI-10/A-Zn-NPI-10, for example, shows peaks at 2962 and 2874 cm^{-1} that are related to the stretching vibration absorption of saturated hydrocarbons (CH_2 , CH_3). Owing to the strong association between the hydroxide radical

and carboxyl group, the peak of the hydroxide radical is shifted to a lower wavenumber of 3000 cm^{-1} , overlapping with the saturated hydrocarbon peaks, and can be identified by ^1H NMR. The stretching vibration absorption of the carbonyl group is characterized by a very intense peak at 1732 cm^{-1} . The peaks at 1249 and 1167 cm^{-1} are assigned to the asymmetric and symmetric C–O–C stretching vibration absorptions, respectively. Free-radical polymerization of acrylic monomers was carried out in the presence of an initiator. The aryl radical at 1534 cm^{-1} and the N–H bond at 3373 cm^{-1} indicate that the indole derivative was copolymerized with acrylic monomers. The peak at approximately 1599 cm^{-1} corresponding to the COO–Zn–COO stretching vibration appears for sample A-Zn-NPI-10, but it was not present in the A-NPI-10 IR spectrum. This result indicates that Zn^{2+} was introduced to A-Zn-NPI-10 on the basis of A-NPI-10.

The changes in structure induced by the polymerization process were further characterized by ^1H NMR. The peaks at 11.1 and 8.4 ppm are attributable to the NH signal of the pyrrole ring and the acylamino group, respectively. The peaks from the unsaturated hydrogen of the indole ring fall in the region from 7 to 8 ppm. The peak at 4.5 ppm is assigned to the methylene hydrogen ($\text{CH}_2\text{-NH-CO}$), which is connected to the acylamino group. These results indicate that the indole derivative was introduced to the polymer. The peak at 12.3 ppm is related to the OH signal of the carboxyl group that disappears after the dehydration condensation reaction between the prepolymer and $\text{Zn}(\text{OH})_2$. The peak at 4 ppm can be assigned to the saturated hydrogen in the side chain of acrylic acid. The signals between 1 and 2 ppm also correspond to the saturated hydrocarbon in the side chain. The absence of peaks at 5.6 and 6.2 ppm indicates that the alkene groups in the reactants participate in the polymerization reaction. The IR and ^1H NMR spectra collectively demonstrate that acrylate resins containing the indole derivative structure were successfully synthesized. The IR and ^1H NMR spectra of the acrylate resins are presented in the supplementary material (Fig. A. 6).

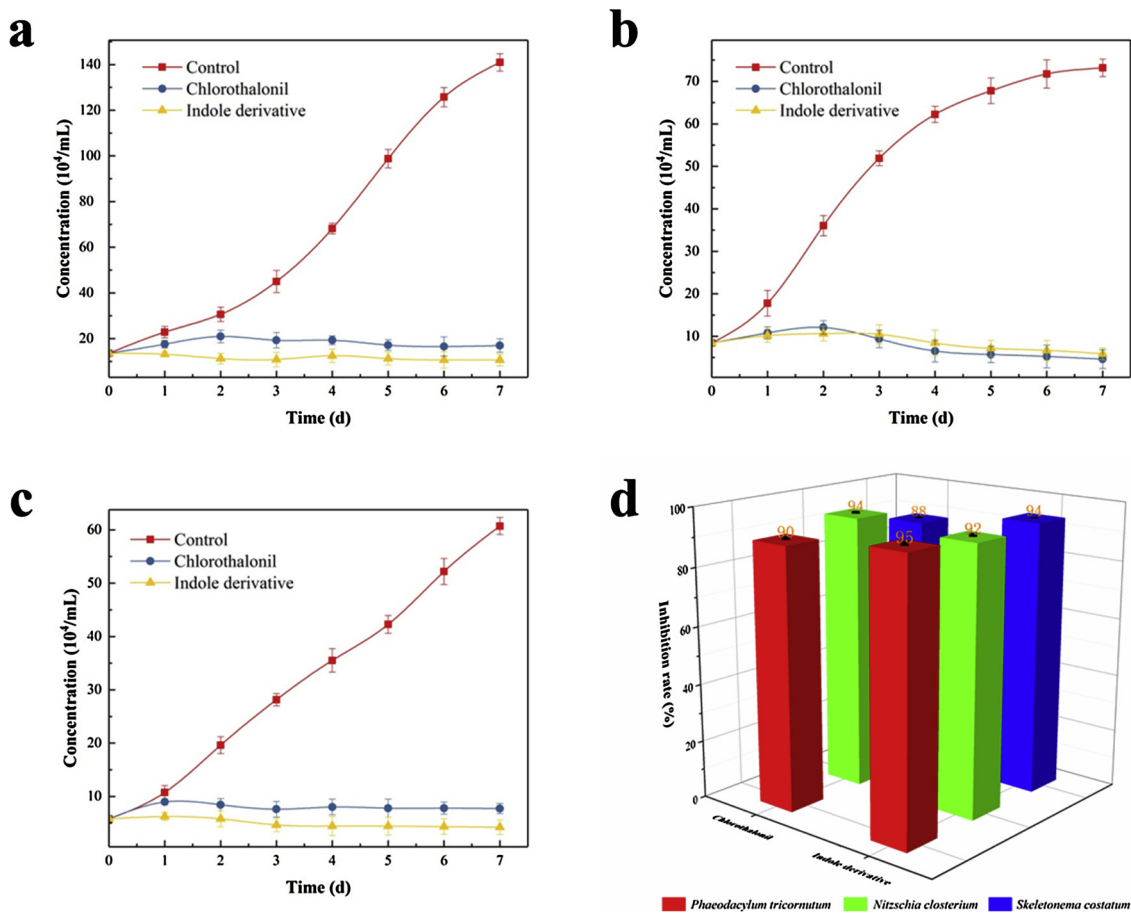


Fig. 3. Growth curves of *Phaeodactylum tricornutum* (a), *Nitzschia closterium* (b) and *Skeletonema costatum* (c); column graph of inhibition rate (d).

3.4. Self-polishing performance analysis

The self-polishing performance of the acrylate resins was evaluated by the abrasion quantity during the dynamic simulation and the static immersion experiments (Fig. 4) by laser scanning confocal microscopy images of surface height variations (Fig. 5). Fig. 4 demonstrates the prominent abrasion quantity in the initial stage of the measurement, which is possibly caused by erosion by seawater of residual solvents, small molecules, etc. The variation in the abrasion quantity becomes gradually level in the subsequent phase of the experiment. The abrasion quantities in the dynamic simulation experiment are distinctly higher than those in the static immersion experiment due to the different

degrees of polishing. Dynamic and static abrasion quantities are affected by the hydrolysis of ester groups; however, the former is also influenced by polishing under the dissolution and scouring of seawater, whereas the latter is less affected by polishing under the dissolution of seawater. Compared with that of pure acrylate resins, the self-polishing rate of the acrylate resins containing indole derivative is reduced, and the greater the amount of indole derivative introduced to the acrylate resins is, the lower the abrasion quantity is. In brief, the self-polishing property is still maintained after introducing the indole derivative.

Fig. 5 reveals laser scanning confocal microscopy images of surface height variations in the process of self-polishing. The surface height was evaluated to determine the surface smoothness of coatings during

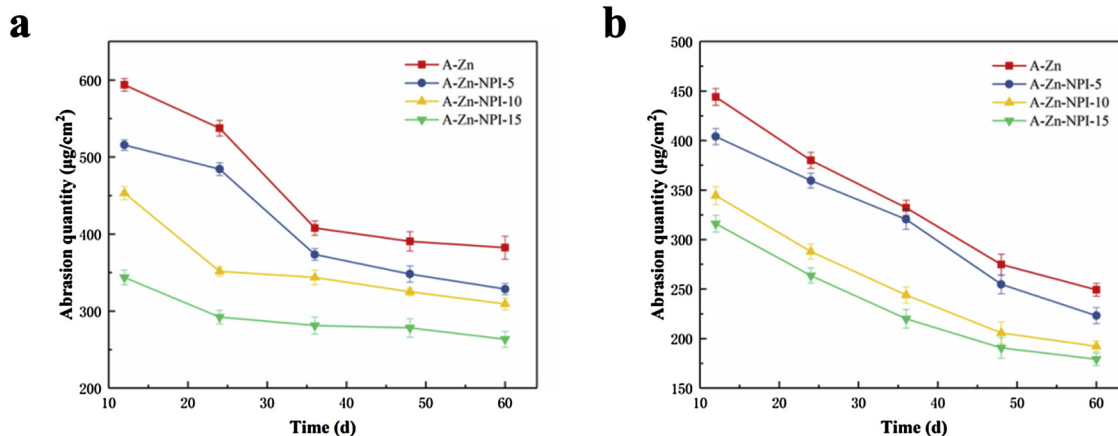


Fig. 4. The abrasion quantity of dynamic simulation (a) and static immersion experiment (b).

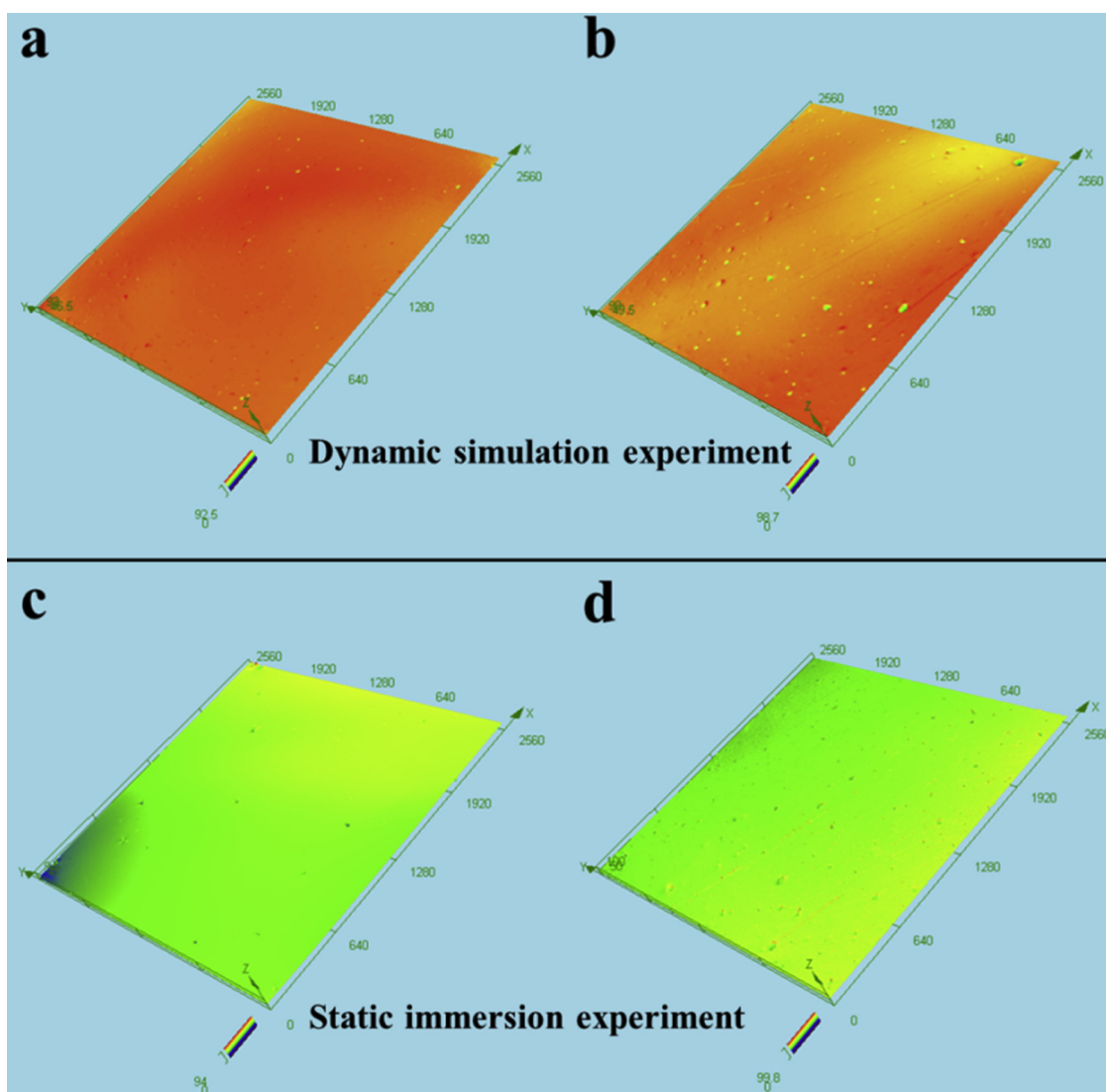


Fig. 5. Laser scanning confocal microscopy images of surface altitude variations before immersion (a), (c), after 5 cycles immersion in seawater (b), (d).

different treatments. The colors of the height variations were nearly the same as those corresponding to the smooth surface before immersion. After the dynamic simulation and static immersion experiments, subsidence holes appeared on the surface of the resins in comparison with the resins in their original state. The attrition degree of the dynamic simulation experiment (Fig. 5b) is higher than that of the static immersion experiment (Fig. 5d). This difference is mainly because the self-polishing property affects the former to a greater extent than the latter. Nevertheless, the surfaces of the coatings were relatively flat after 5 cycles. The changes in abrasion quantity and surface height indicate that the surface of acrylate resins containing the indole derivative was polished steadily, which could guarantee persistent self-polishing behavior.

3.5. Algal inhibition test of acrylate resins

The algae inhibition rate of the resins is shown in Fig. 6. The anti-algal properties of the resins containing the indole derivative are obviously better than those of the resins without the indole derivative. Comparing the A-Zn samples with the A-Zn-NPI-x samples ($x = 5, 10, 15$) revealed that the inhibition rates of A-Zn-NPI-x were improved by one fold. The greater the amount of more indole derivative introduced to the resins was, the higher the corresponding inhibition rates were.

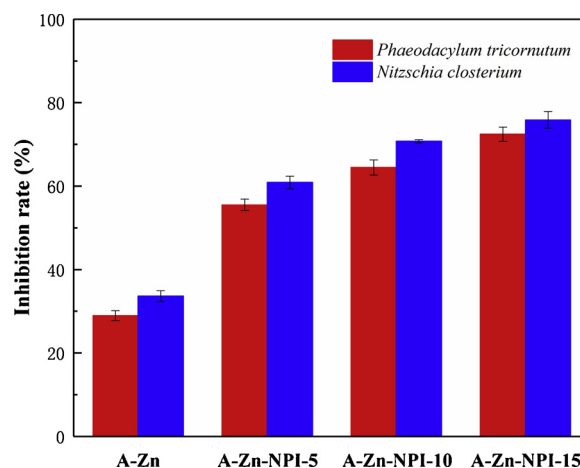


Fig. 6. The algal inhibition rates of resins: *Phaeodactylum tricornutum* (a), *Nitzschia closterium* (b).

Furthermore, the results reveal that the algae inhibition rates against *Nitzschia closterium* are slightly higher than those on *Phaeodactylum tricornutum*. In the present study, the algal inhibition test was

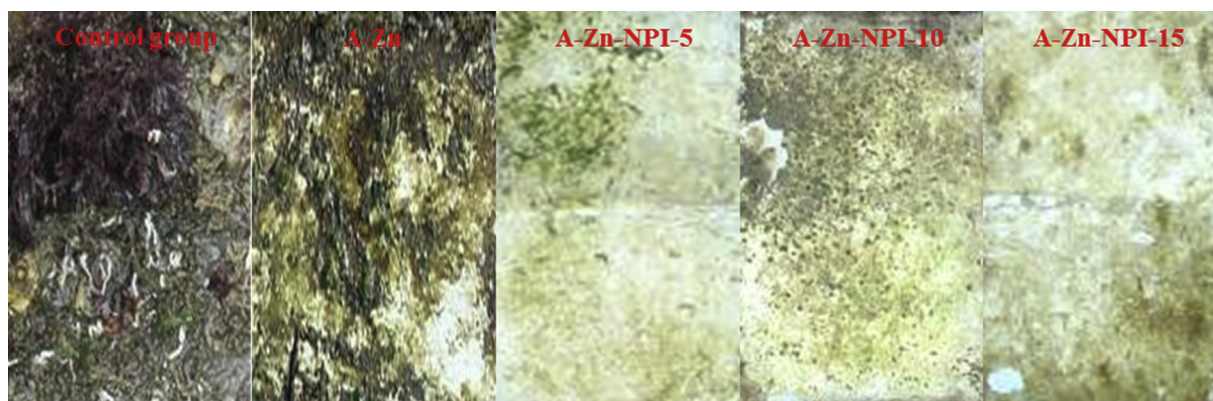


Fig. 7. Images of coatings after immersion in the sea for three months.

performed in a static environment less affected by polishing, so the difference in the quantity of the indole derivative played a critical role in the test of the acrylate resins. The greater the amount of more indole derivative introduced to the acrylate resins was, the higher the algae inhibition rate was. The reason for this trend was the ability of the indole derivative introduced to the acrylate resins to interfere with the equilibrium of calcium ions in algal cells to inhibit the marine algae growth. These results indicate that acrylate resins containing the indole derivative could be highly effective in inhibiting biofouling.

3.6. Antifouling test in marine

Images of various coatings after immersion in the sea for three months are included in Fig. 7. There are distinct differences between coatings containing different contents of the indole derivative. The surface of the coatings in the control group was fully covered by algae. In contrast, there were few fouling organisms present on the A-Zn-NPI-10 and A-Zn-NPI-15 samples. The greater the amount of indole derivative introduced to the coatings was, the better the antifouling properties of the coatings were. There are two reasons for this phenomenon: the indole derivative itself and self-polishing property. First, the presence of the indole derivative suspending the side chain could make it difficult for fouling organisms to grow. The inhibition mechanism against marine algae growth is speculated to be the indole derivative interfering with the equilibrium of calcium ions in algal cells and decreasing the abundance of cellular Ca^{2+} [37,40]. This mechanism is effective because calcium ions participate in nearly all algae physiological activities and are closely related to the growth and development of microorganisms [39]. Second, the self-polishing performance of acrylate resins can make a surface renew continuously by hydrolysis of the ester groups under the influence of seawater. The self-polishing performance could make attached organisms readily fall off the surface and exposed additional indole derivative structures on the new surface. Therefore, the combination of these two functions of the resins can significantly improve their antifouling properties.

4. Conclusions

The novel indole derivative was synthesized via a Friedel-Crafts alkylation reaction. This derivative showed the high biological activities and thus was introduced to the side chain of acrylate resins. Self-polishing experiments showed that the acrylate resins to which the indole derivative was introduced still exhibited prominent polishing properties. Furthermore, biological simulation tests revealed that the acrylate resins suspending the indole derivative in the side chain displayed better antifouling performance than the acrylate resins without the indole derivative (A-Zn). The above experiments collectively indicated that the antifouling property of the acrylate resins was significantly promoted by the combination of the biofouling resistance of

the indole derivative and the self-polishing properties of acrylate. The results were also verified in a marine antifouling test. This work lays an academic foundation for studying environmentally friendly and highly efficient antifouling coatings.

Declaration of Competing Interest

None.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.colsurfb.2019.110518>.

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